

Phase 1 Verification and Validation

Project / System
Elevator System

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Version
1.2

Date
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Approved By
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Phase 1 incorporates the general operation of the elevator platform. Equipped on the elevator are multiple STM32F303RE development boards, with customized nucleo shield boards that have three buttons on them. There is an LED associated with each pushbutton, as well as three LEDs in the top left that are used to indicate the current floor of the elevator. The elevator controller set up is not modified in this offering of projects, and has been omitted from the test plan. The following items line up with the MuSCoW requirements agreed on by the group.

Summary									
Total Tests	Passed	Failed	Blocked	Not Run	Completion %				
20	15	1	0	0	75%				
ID	Category	Sub Item	Verification Method	Expected Result	Test Procedure	Acceptance Criteria	Result	Validator / Date	Evidence / Notes
P1-VV-001	Floor Node 1	Calls Car	Functional test / observation	Calls Car performs as intended for Floor Node 1.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	Pass	LE/06-19-26	
P1-VV-002	Floor Node 1	Request LED	Functional test / observation	Request LED performs as intended for Floor Node 1.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	Fail		As of the 19th, indicator would flick on but not turn off.
P1-VV-003	Floor Node 1	Floor Indicator	Functional test / observation	Floor Indicator performs as intended for Floor Node 1.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	Pass	LE/06-19-26	
P1-VV-004	Floor Node 1	Door open/close function	Functional test / observation	Door open/close function performs as intended for Floor Node 1.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	N/A		Not implemented in this phase
P1-VV-005	Floor Node 2	Calls Car	Functional test / observation	Calls Car performs as intended for Floor Node 2.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	Pass	LE/06-19-26	
P1-VV-006	Floor Node 2	Request LED	Functional test / observation	Request LED performs as intended for Floor Node 2.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	Pass	LE/06-19-26	
P1-VV-007	Floor Node 2	Floor Indicator	Functional test / observation	Floor Indicator performs as intended for Floor Node 2.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	Pass	LE/06-19-26	
P1-VV-008	Floor Node 2	Door open/close function	Functional test / observation	Door open/close function performs as intended for Floor Node 2.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	N/A		Not implemented in this phase
P1-VV-009	Floor Node 3	Calls Car	Functional test / observation	Calls Car performs as intended for Floor Node 3.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	Pass	LE/06-19-26	
P1-VV-010	Floor Node 3	Request LED	Functional test / observation	Request LED performs as intended for Floor Node 3.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	Pass	LE/06-19-26	
P1-VV-011	Floor Node 3	Floor Indicator	Functional test / observation	Floor Indicator performs as intended for Floor Node 3.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	Pass	LE/06-19-26	
P1-VV-012	Floor Node 3	Door open/close function	Functional test / observation	Door open/close function performs as intended for Floor Node 3.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	N/A		Not implemented in this phase
P1-VV-013	Car Controller	Calls Car	Functional test / observation	Calls Car performs as intended for Car Controller.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	Pass	LE/06-19-26	
P1-VV-014	Car Controller	Request LED	Functional test / observation	Request LED performs as intended for Car Controller.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	Pass	LE/06-19-26	
P1-VV-015	Car Controller	Floor Indicator	Functional test / observation	Floor Indicator performs as intended for Car Controller.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	Pass	LE/06-19-26	
P1-VV-016	Car Controller	Door open/close function	Functional test / observation	Door open/close function performs as intended for Car Controller.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	N/A		Not implemented in this phase
P1-VV-017	Supervisor Controller	Calls Car	Functional test / observation	Calls Car performs as intended for Supervisor Controller.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	Pass	LE/06-19-26	Car called from browser on raspberry pi
P1-VV-018	Supervisor Controller	Request Indicator	Functional test / observation	Request Indicator performs as intended for Supervisor Controller.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	Pass	LE/06-19-26	Requested floor shown in browser option
P1-VV-019	Supervisor Controller	Received Message	Functional test / observation	Received Message performs as intended for Supervisor Controller.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	Pass	LE/06-19-26	Controller receives and sends messages as intended
P1-VV-020	Supervisor Controller	CAN Bus information	Functional test / observation	CAN Bus information performs as intended for Supervisor Controller.	1) Initiate test condition. 2) Observe response. 3) Record result and evidence.	Pass when observed response matches expected behaviour without errors.	Pass	LE/06-19-26	Simple display printout is available at terminal.